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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

# TAPE, LACING AND TYING, GLASS

The General Services Administration has authorized the use of this commercial item description as a replacement for Type IV of MIL-T-43435B for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers glass tape, flat braided, for lacing and tying.

2. CLASSIFICATION. Tape shall conform to the sizes and finishes as specified. (see 3.3 and 3.6)

2.1 Size. The size shall be identified as 1,2,3,4 or 5. (see Table 1)

2.2 Finish. The finish shall be identified as C, D or F. (see 3.6)

3. SALIENT CHARACTERISTICS.

3.1 Material. The yarn used in the fabrication of the tape shall be electrical grade, continuous filament, glass yarn having high insulation resistance, high dielectric strength, high resistance to aging and low moisture pick-up. The material shall be free from any alkali metal oxides, such as soda or potash, and from foreign particles, dirt, or other impurities.

3.2 Construction. The tape shall be a flat braid construction.

3.3 Physical Requirements. Tape shall conform to the requirements of Table 1.

TABLE 1: Tape, Lacing and Tying, Glass

| Size | Width in | | Thickness in Inches | | Breaking |
|---|---|---|---|---|---|
| | Min | Max | Min | Max | (Lbs, min) |
| 1 | .203 | .248 | .013 | .019 | 200 |
| 2 | .099 | .121 | .013 | .019 | 100 |
| 3 | .077 | .094 | .013 | .019 | 75 |
| 4 | .054 | .066 | .013 | .019 | 50 |
| 5 | .045 | .055 | .013 | .019 | -- |

3.4 Elongation. Maximum elongation at breaking strength shall not be greater than 5% when tested as specified in FED-STD-191, Method 4108. This test shall be performed simultaneously with breaking strength. Elongation for size 5 is not rated.

3.5 Color. The color of this tape shall be natural. No other colors are available.

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3.6 Finish. A finish option shall be specified, in accordance with 3.6.1 through 3.6.3. Tape finishes shall not contain any mercury or copper. All finishes shall be able to be used freely in direct contact with insulated cable or wire.

3.6.1 Finish C - Synthetic rubber. Tape shall be uniformly impregnated with a synthetic rubber or elastomer finish. Treated tape shall contain 7-17% rubber by weight. (see 5.2.1)

3.6.2 Finish D - TFE-Fluorocarbon coating. Tape shall be uniformly impregnated before braiding. Treated tape shall contain 10-20% TFE-fluorocarbon by weight. (see 5.2.1)

3.6.3 Finish F - Silicone resin impregnated. Tape shall be uniformly impregnated, and shall contain 7-17% silicone resin material by weight. (see 5.2.1)

3.7 Knot Slip Resistance. (Finish C and F only). Stress applied to a specimen made by joining two ends of the braided tape with a square knot shall result in the breakage of the tape specimen rather than the slippage or pulling out of the knot to the point of separation when tested as specified in 5.2.2. If application necessitates the need for Finish D, an additional coat of synthetic elastomer may be added when knot holding characteristics are desired.

3.8 Put-up. Unless otherwise specified, the tape shall be furnished on parallel wind spools (reels) or universal wind tubes (holders). Size 1 and 2 tapes shall be put-up in a minimum of 250 yard lengths, and size 3, 4, and 5 tapes shall be put-up in a minimum of 500 yard lengths. The tape shall be free from twists, lumps, or projecting ends and shall be evenly wound so that each turn and layer is free from entanglement and twisting. There shall be no more than four pieces per reel or holder, and no piece shall be less than 50 yards in length.

3.9 Breaking Strength. Minimum breaking strength shall be as stated in Table 1 when tested as specified in FED-STD-191, Method 4108.

3.10 Fungus Resistance. No tape shall show visible growth (to the naked eye) of fungus on the surface of the test specimens when tested as specified in FED-STD-191, Method 5760.

3.11 Blocking. There shall be no visible damage or removal of the coatings on Finish C, D and F, tapes when tested as specified in paragraph 5.2.3.

3.12 Accelerated Aging. When specified (see 7.4), Finish C, D and F tapes shall show no evidence of stiffness, brittleness, softness, or tackiness when tested as specified in FED-STD-191, Method 5852, except that the specimen shall be six inches in length by the full width of the tape.

3.13 Identification. Tape shall be tagged with a label or ticket containing at least the following information: CID part number, National Stock Number, date of manufacture, and manufacturer's name. When put-up or color is non-standard, the tag shall indicate the actual color or length.

3.14 Flammability. (Finish C and D only) Any flame of the tape shall be self extinguishing within a maximum of 5 seconds after removal of the external flame source from the tape when tested as specified in paragraph 5.2.4.

4. REGULATORY REQUIREMENTS.

4.1 The offerer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).



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## 5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producers own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Inspection Requirements.

5.2.1 Determination of Percentage of Finish. The percentage of finish C, D and F impregnating materials on the respective type tapes shall be determined during processing, by the manufacturer, by weighing an identical sample of tape or yarn (see 3.6), before and after the impregnating process. The test specimen shall be a minimum of 30 yards of tape or 120 yards of yarn as applicable. Weight (Length per pound) shall be determined in accordance with FED-STD-191, Method 4010. The percentage of finish shall be calculated as follows:

$$\text{Percent Finish} \quad \frac{A-B}{B} \times 100$$

Where: A = Length per pound of untreated tape or yarn, as applicable.
B = Length per pound of treated tape or yarn, as applicable.

5.2.2 Determination of Knot Slip Resistance. The knot slip resistance of the tapes shall be determined in accordance with FED-STD-191, Method 4108, except as modified herein. The test specimens shall be cut in half and then the two halves shall be firmly tied together with a square knot. The two protruding ends at the knot shall be cut so that one-half inch protrudes. The specimen shall be aligned so that the knot is approximately halfway between the clamps. The machine shall be operated until the knot slips or pulls out or the specimen breaks at or in proximity to the knot. Breaks closer than 1/2 inch of the jaws shall be discarded. Five good readings shall be obtained. If knot slippage is obtained on only one of five readings, an additional specimen shall be tested and if a good reading is obtained, the reading indicating knot slippage shall be discarded.

5.2.3 Determination of Blocking. Ten turns of the tape shall be wound on a one-quarter inch diameter clean metal mandrel under a two pound tension and the end secured through holes in the mandrel. Eight turns of tape shall then be wound on top of the first layer under the two pound tension and the ends secured as before. The wound mandrel shall then be placed in a temperature controlled oven at $70 \pm 1^{o}$ C for two hours, and in such a manner that no part of the specimen comes in contact with the surface of the oven. The specimen shall then be removed and cooled at room temperature. After cooling, the outer layer shall be unwound and examined for evidence of damage to the coatings due to adhesion between layers of turns. The first layer shall be examined while still in place for similar evidence of adherence and damage.

5.2.4 Determination of Flammability. (Finish C and D only) The flammability testing shall be performed in accordance with the following test method:

5.2.4.1 Apparatus. The flammability test chamber shall be approximately 1 foot square by 2 feet in height, and shall be open at top and front to provide adequate ventilation for combustion but prevent drafts. Means shall be provided in the chamber to hold the test specimen taut at an angle of 60 degrees from horizontal in a vertical plane parallel to and about 6 inches in front of the rear wall of the chamber. The test burner shall be a Bunsen type gas burner having a 1/4 inch inlet, a needle valve in the base for gas adjustment, a bore of 3/8 inch nominal, and a length of approximately 4 inches above the air inlets. The burner shall be fitted with a wing top flame spreader having a 1/16 inch by 2 inch opening.



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5.2.4.2 Procedure. A 24-inch specimen of tape, marked at a distance of 8 inches from the lower end to indicate the point of contact for the test flame, shall be clamped tautly, at 60 degrees from horizontal, in the specimen holder of the test chamber. The burner shall be adjusted to deliver an all-blue flame, 2 inches high, at a temperature of 955 °C ± 30 °C (1751 °F ± 54 °F), as measured with a thermocouple pyrometer. With the burner held at a 90 degree angle to the tape specimen and the long dimension of the flame spreader parallel to the axis of the specimen, the hot tip of the flame shall be applied to the tape so that the midpoint of the flame touches the 8-inch mark on the specimen. The period of flame application shall be 30 seconds. At the close of the application period, the flame shall be withdrawn and the duration of the after flame and the burned length of the specimen shall be measured and recorded.

6. PACKAGING.

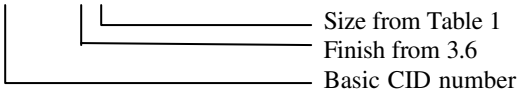
6.1 Preservation, packing and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Intended use. The tape is intended for lacing and tying electrical wire and cable bundles. The application of an additional finish to glass tape with Finish D may be used when better knot holding characteristics are desired. Tape with Finish C provides the best knot holding characteristics. Useful temperature range of glass tape is -100 F to 800 F.

7.2 Part Identification Number (PIN). The PIN shall consist of the basic CID number, followed by the finish code letter from 3.5, followed by the size number from Table 1.

PIN Example: AA52083- A-2




7.3 Source of Documents.

7.3.1 Federal Standards. Federal Standards are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

7.3.2 FAR. The FAR may be obtained from the Regulatory Secretariat, 1800 F Street NW, Washington DC 20405 or online at http://www.acqnet.gov.

7.4 Ordering Data.

a. Title, number and date of this Commercial Item Description.
b. Size and finish required. (see 3.3 and 3.6)
c. Color, if other than specified. (see 3.5)
d. Put-up if other than specified. (see 3.8)
e. When accelerated aging is required. (see 3.12)
f. Selection of applicable levels of packaging and packing. (see 6)

7.5 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. This list may not be indicative of all possible NSNs associated with the CID.

| NSN | Type | Size | Finish | Color |
|---|---|---|---|---|
| 4020-00-823-6997 | 4 | 1 | D | Natural |
| 4020-00-100-9067 | 4 | 2 | D | Natural |
| 4020-01-354-7284 | 4 | 2 | F | Natural |
| 4020-01-411-2548 | 4 | 3 | C | Natural |
| 4020-00-492-3975 | 4 | 3 | D | Natural |
| 4020-00-053-5738 | 4 | 4 | D | Natural |
| 4020-00-123-8675 | 4 | 5 | D | Natural |



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7.6 Key words.

Silicone resin impregnated
Synthetic rubber
TFE Fluorocarbon coating

7.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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